

ASSIGNMENT-4.2

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Batch-37

Task Description-1

- Zero-shot: Prompt AI with only the instruction. Write a Python function to determine whether a given number is prime

Expected Output-1

- A basic Python function to check if a number is prime, demonstrating correct logical conditions without relying on examples or additional context

Prompt:-

Using python function to whether a given number is prime

Code:-

```
#Using python function to whether a given number is prime
def is_prime(number):
    """Check if a number is prime.

    A prime number is a natural number greater than 1 that cannot be
    formed by multiplying two smaller natural numbers.

    Usage example:
        is_prime(7)  # Returns True
        is_prime(10) # Returns False
    """
    if number <= 1:
        return False
    for i in range(2, int(number**0.5) + 1):
        if number % i == 0:
            return False
```

```

    return True
if __name__ == '__main__':
    num = int(input("Enter a number to check if it is prime: "))
    if is_prime(num):
        print(f"{num} is a prime number.")
    else:
        print(f"{num} is not a prime number.")

```

Explain code:-

This Python program checks whether a given number is a prime number or not using a function.

Task Description-2

- One-shot: Provide one example: Input: [1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements in a list.

Expected Output-2

- A correct conversion function guided by the single example.

Prompt:-

Using python function calculate the sum of elements in a list

#example :-Input: [1, 2, 3, 4]

#output: 10

Code:-

```

#Using python function calculate the sum of elements in a list
#example :-Input: [1, 2, 3, 4]
#output: 10
def sum_of_elements(elements):
    """Calculate the sum of elements in a list.

    Args:
        elements (list): A list of numeric elements.

```

```

Returns:
    float: The sum of the elements in the list.

Usage example:
    sum_of_elements([1, 2, 3, 4]) # Returns 10
    sum_of_elements([5.5, 4.5])  # Returns 10.0
"""
total = 0
for element in elements:
    total += element
return total
if __name__ == '__main__':
    input_list = input("Enter a list of numbers separated by spaces: ")
    try:
        elements = [float(x) for x in input_list.split()]
        result = sum_of_elements(elements)
        print(f"The sum of the elements is: {result}")
    except ValueError:
        print("Error: Please enter valid numeric values.")

```

Explain code:-

The program takes a list of numbers from the user and calculates their sum using a function. It adds each element one by one and displays the final total.

Task Description-3

- Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

Expected Output-3

- Accurate function that returns only the digits from alphanumeric string.

Prompt:-

Task: Create a Python function that extracts only digits from an alphanumeric string.

#Example 1:

#Input: "a1b2c3"

#Output: "123"

#Example 2:

#Input: "x9y8z7"

#Output: "987"

#Example 3:

#Input: "abc123xyz"

#Output: "123"

Code:-

```
#Task: Create a Python function that extracts only digits from an
alphanumeric string.
#Example 1:
#Input: "a1b2c3"
#Output: "123"
#Example 2:
#Input: "x9y8z7"
#Output: "987"

#Example 3:
#Input: "abc123xyz"
#Output: "123"
def extract_digits(input_string):
    """Extracts and returns only the digits from an alphanumeric string.

    Args:
        input_string (str): The alphanumeric string to extract digits
        from.

    Returns:
        str: A string containing only the digits extracted from the input
        string.
```

```

Usage example:
    extract_digits("a1b2c3")      # Returns "123"
    extract_digits("x9y8z7")      # Returns "987"
    extract_digits("abc123xyz")    # Returns "123"

"""
digits = ''.join([char for char in input_string if char.isdigit()])
return digits
if __name__ == '__main__':
    user_input = input("Enter an alphanumeric string: ")
    result = extract_digits(user_input)
    print(f"Extracted digits: {result}")

```

Explain code:-

This program defines a function that goes through each character of an alphanumeric string and collects only the digits.

It uses `isdigit()` to check whether a character is a number and joins all such characters into a new string.

The user enters a string, and the program prints only the extracted digits.

Task Description-4

- Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

Expected Output-4

- Output comparison + student explanation on how examples helped the model.

Prompt:

Zero-shot:- Write a Python function that counts the number of vowels in a given string.

Few-shot:-

Task: Create a Python function that counts the number of vowels in a string.

Example 1:

Input: "hello"

Output: 2

Example 2:

Input: "education"

Output: 5

Code:-

```
#Write a Python function that counts the number of vowels in a given string.
```

```
def count_vowels(input_string):  
  
    vowels = "aeiouAEIOU"  
  
    count = sum(1 for char in input_string if char in vowels)  
  
    return count  
  
if __name__ == '__main__':  
  
    user_input = input("Enter a string: ")  
  
    result = count_vowels(user_input)  
  
    print(f"Number of vowels in the given string: {result}")
```

```
"""Task: Create a Python function that counts the number of vowels in a string.
```

```
Example 1:
```

```
Input: "hello"
```

```
Output: 2
```

Example 2:

Input: "education"

Output: 5

Now write the function."""

```
def count_vowels(input_string):  
  
    vowels = "aeiouAEIOU"  
  
    count = sum(1 for char in input_string if char in vowels)  
  
    return count  
  
if __name__ == '__main__':  
  
    user_input = input("Enter a string: ")  
  
    result = count_vowels(user_input)  
  
    print(f"Number of vowels in the given string: {result}")
```

Explain code:-

In few-shot prompting, examples like "hello" → 2 and "education" → 5 are provided first, which helps the model clearly understand what to count before writing the vowel-counting function.

In zero-shot prompting, no examples are given, so the model relies only on the instruction, which works for simple tasks but offers less guidance than few-shot prompting.

Task Description-5

- Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using the built-in min() function.

Expected Output-5

- A function that handles all cases with correct logic based on example patterns.

Prompt:-

Create a Python function that finds the minimum of three numbers without using the built-in min() function .

Example 1:

Input: a = 3, b = 7, c = 5

Output: 3

Example 2:

Input: a = 10, b = 2, c = 8

Output: 2

Example 3:

Input: a = -1, b = -5, c = 0

Output: -5

Code:-

```
"""Create a Python function that finds the minimum of three numbers  
without using the built-in min() function Using python function.
```

```
Example 1:
```

```
Input: a = 3, b = 7, c = 5
```

```
Output: 3
```

```
Example 2:
```

```
Input: a = 10, b = 2, c = 8
```

Output: 2

Example 3:

Input: a = -1, b = -5, c = 0

Output: -5"""

```
def find_minimum(a, b, c):  
    """Find the minimum of three numbers without using the built-in min()  
function.  
  
    Args:  
  
        a (float): The first number.  
  
        b (float): The second number.  
  
        c (float): The third number.  
  
    Returns:  
  
        float: The minimum of the three numbers.  
  
    Usage example:  
  
        find_minimum(3, 7, 5)    # Returns 3  
  
        find_minimum(10, 2, 8)   # Returns 2  
  
        find_minimum(-1, -5, 0)  # Returns -5  
  
    """  
  
    if a <= b and a <= c:  
  
        return a  
  
    elif b <= a and b <= c:  
  
        return b  
  
    else:
```

```
        return c

if __name__ == '__main__':

    try:

        a = float(input("Enter the first number (a): "))

        b = float(input("Enter the second number (b): "))

        c = float(input("Enter the third number (c): "))

        result = find_minimum(a, b, c)

        print(f"The minimum of the three numbers is: {result}")

    except ValueError:

        print("Error: Please enter valid numeric values.")
```

Explain code:-

This program defines a function that compares three numbers using conditional statements to find the smallest one without using the built-in `min()` function.

The user enters three numbers, the function checks all cases logically, and then prints the minimum value.